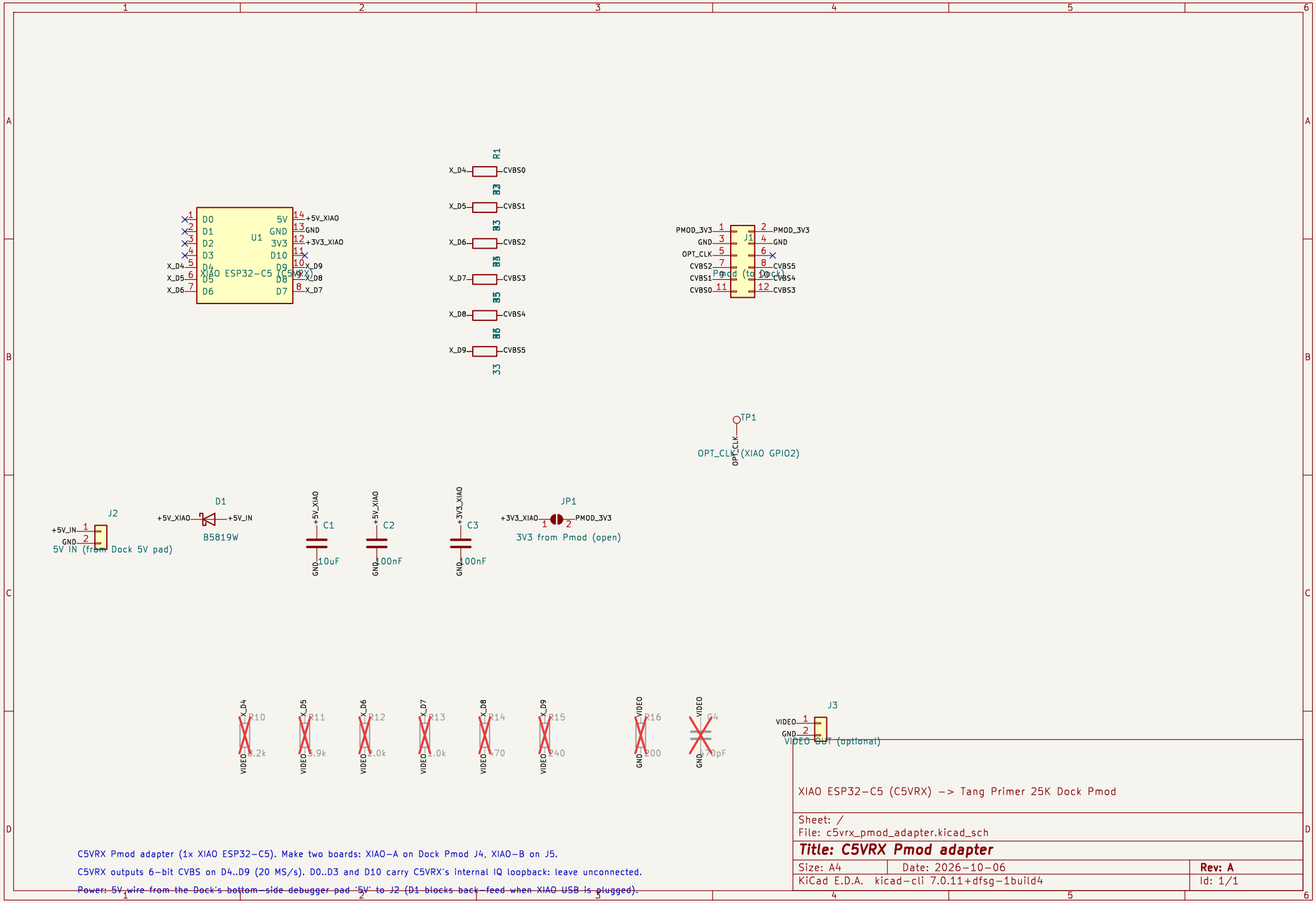

U1
XIAO ESP32-C5 (C5VRX)
D0
D1
D2
D3
D4
D5
D6
5V
GND
3V3
D10
D9
D8
D7
+5V_XIAO
GND
+3V3_XIAO
X_D9
X_D8
X_D7
R1
X_D4
X_D5
X_D6
X_D7
X_D8
X_D9
CVBS0
CVBS1
CVBS2
CVBS3
CVBS4
CVBS5
J1
Pmod (to Dock)
PMOD_3V3
GND
OPT_CLK
CVBS2
CVBS1
CVBS0
CVBS5
CVBS4
CVBS3
TP1
OPT_CLK (XIAO GPIO2)
J2
5V IN (from Dock 5V pad)
+5V_IN
GND
D1
B5819W
C1
10uF
C2
100nF
C3
100nF
JP1
3V3 from Pmod (open)
+3V3_XIAO
PMOD_3V3
R10
3.2k
R11
3.9k
R12
2.0k
R13
1.0k
R14
470
R15
240
R16
200
C4
470pF
VIDEO
J3
VIDEO OUT (optional)
XIAO ESP32-C5 (C5VRX) -> Tang Primer 25K Dock Pmod
Sheet: /
File: c5vrx_pmod_adapter.kicad_sch
Title: C5VRX Pmod adapter
Size: A4
Date: 2026-10-06
Rev: A
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4
Id: 1/1
C5VRX Pmod adapter (1x XIAO ESP32-C5). Make two boards: XIAO-A on Dock Pmod J4, XIAO-B on J5.
C5VRX outputs 6-bit CVBS on D4..D9 (20 MS/s). D0..D3 and D10 carry C5VRX's internal IQ loopback: leave unconnected.
Power: 5V wire from the Dock's bottom-side debugger pad '5V' to J2 (D1 blocks back-feed when XIAO USB is plugged).
JP1: close only to run from Pmod 3V3 instead of J2 (then never plug the XIAO USB while JP1 is closed).
R10..R16, C4: optional analog CVBS DAC (C5VRX reference values), not populated by default.